Lab – Forensic Analysis of Malware Using Wireshark

Overview

In the realm of network forensics, Wireshark stands out as a powerful tool, particularly in uncovering the intricacies of malware. Through Packet Capture (PCAP) traffic analysis, Wireshark allows analysts to delve into the behavior of potential threats, shedding light on their patterns, origins, and impact. In this short lab, we will analyze the Dridex malware.

Dridex is the name for a family of information-stealing malware that has also been described as a banking Trojan. This malware first appeared in 2014 and has been active ever since.

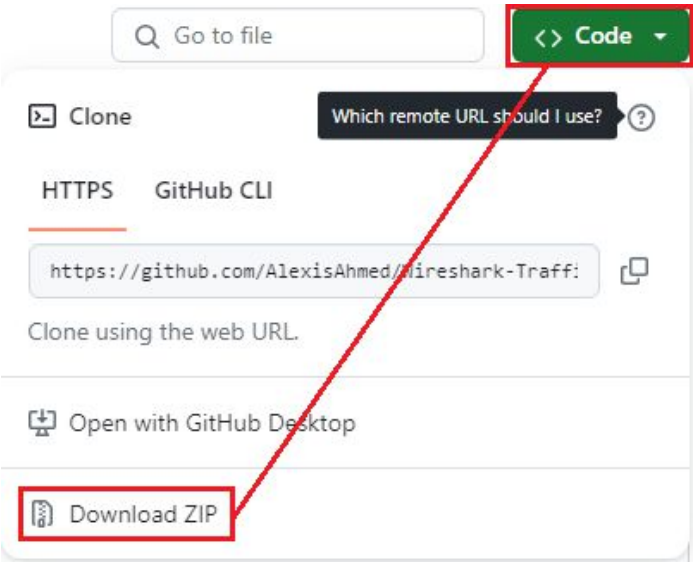
This lab will explore how Wireshark can provide critical insights crucial for threat mitigation and system defense.

Warning: The pcap used for this tutorial contains Windows-based malware. This lab was performed using a virtual install of Kali Linux. It is recommended that the lab be performed on a non-windows platform.

Lab Requirements

- One installation of Kali Linux
- One Download of the Wireshark-Traffic-Analysis-main.zip file. The archive is password protected; to decrypt it, use the password "infected."

Download the needed files from GitHub. Expand the green code button and use the Download zip option to start the download.

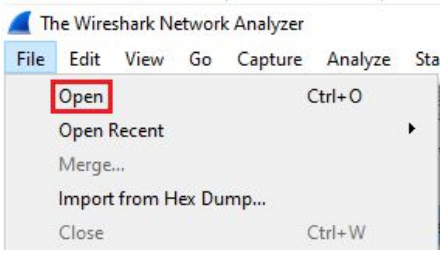


Extract the files in the archive using the password “infected.”

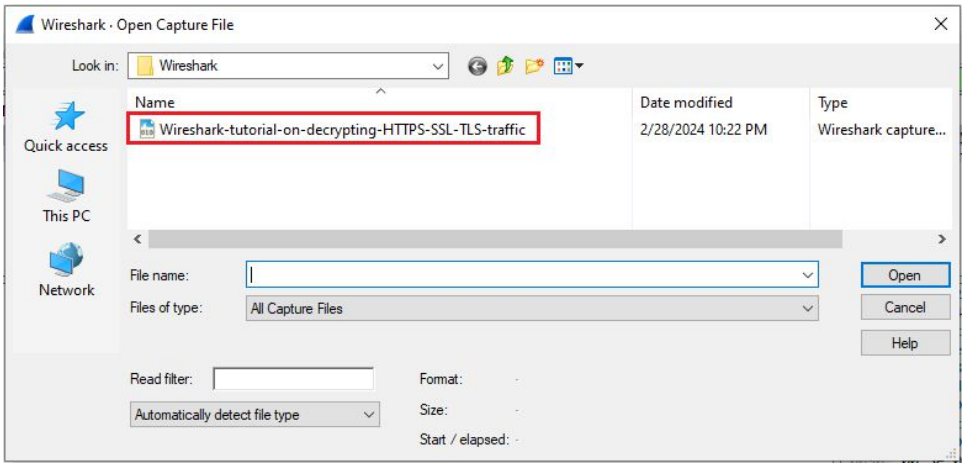
Remember where y0u saved the two extracted files. You will need both.

Import the pcap into Wireshark.

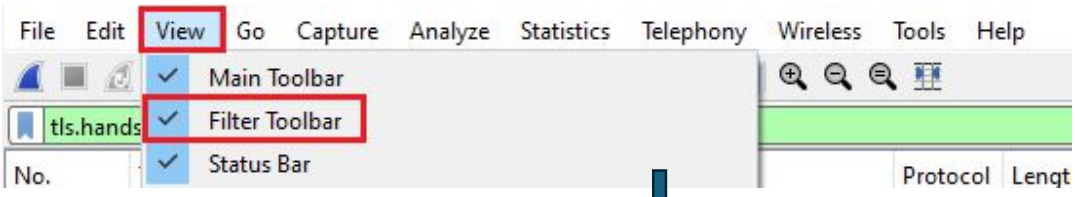
Open Wireshark. From the taskbar, click, file and then open.



Browse to the location of the extracted files and x2click on the pcap file shown.



With the pcap loaded, we can begin by filtering for successful TLS handshakes by typing the following filter in the filter toolbar. If you do not see the filter toolbar, from the Wireshark taskbar, click View and check the option to show the filter toolbar.




```
tls.handshake.type eq 1
```

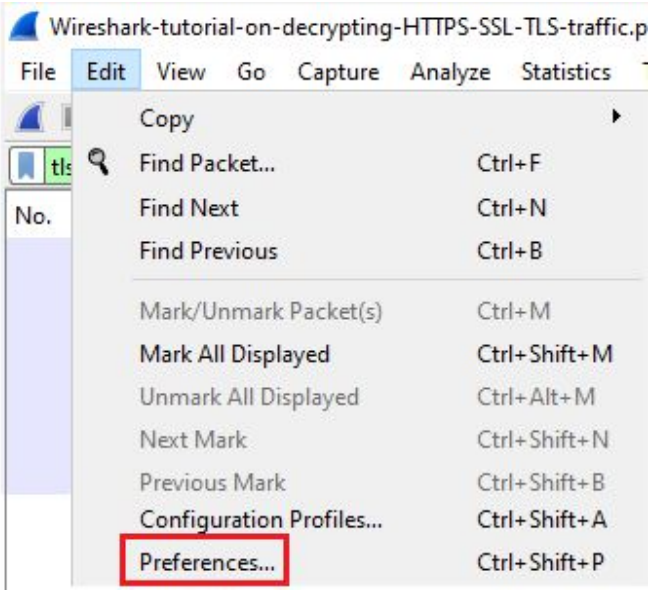
tls.handshake.type eq 1

No.	Time	Source	Source Port	Destination	Destination Port	Protocol	Length	Info
60	17.416904	10.4.1.101	49877	13.107.3.128	443	TLSv…	240	Client Hello
121	27.938662	10.4.1.101	50051	40.122.160.14	443	TLSv…	236	Client Hello
157	29.446603	10.4.1.101	50074	94.103.84.245	443	TLSv…	240	Client Hello
820	128.832771	10.4.1.101	51649	40.122.160.14	443	TLSv…	428	Client Hello
1008	273.354367	10.4.1.101	53857	20.191.48.196	443	TLSv…	272	Client Hello
1236	565.816488	10.4.1.101	58338	162.255.119.253	443	TLSv…	216	Client Hello
1328	696.284167	10.4.1.101	60335	162.255.119.253	443	TLSv…	392	Client Hello
1420	832.344860	10.4.1.101	62415	162.255.119.253	443	TLSv…	392	Client Hello

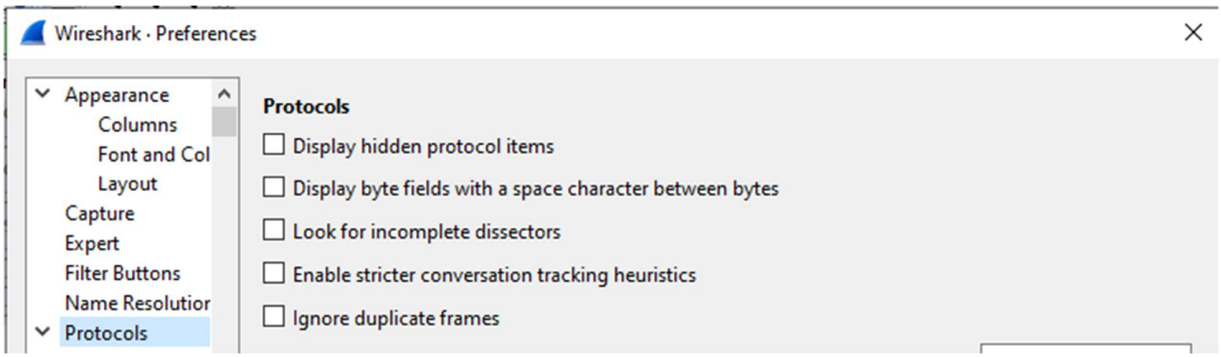
Decrypt the TLS traffic.

Since the traffic is encrypted, we will need to decrypt it using the decryption key provided in the downloaded archive from GitHub.

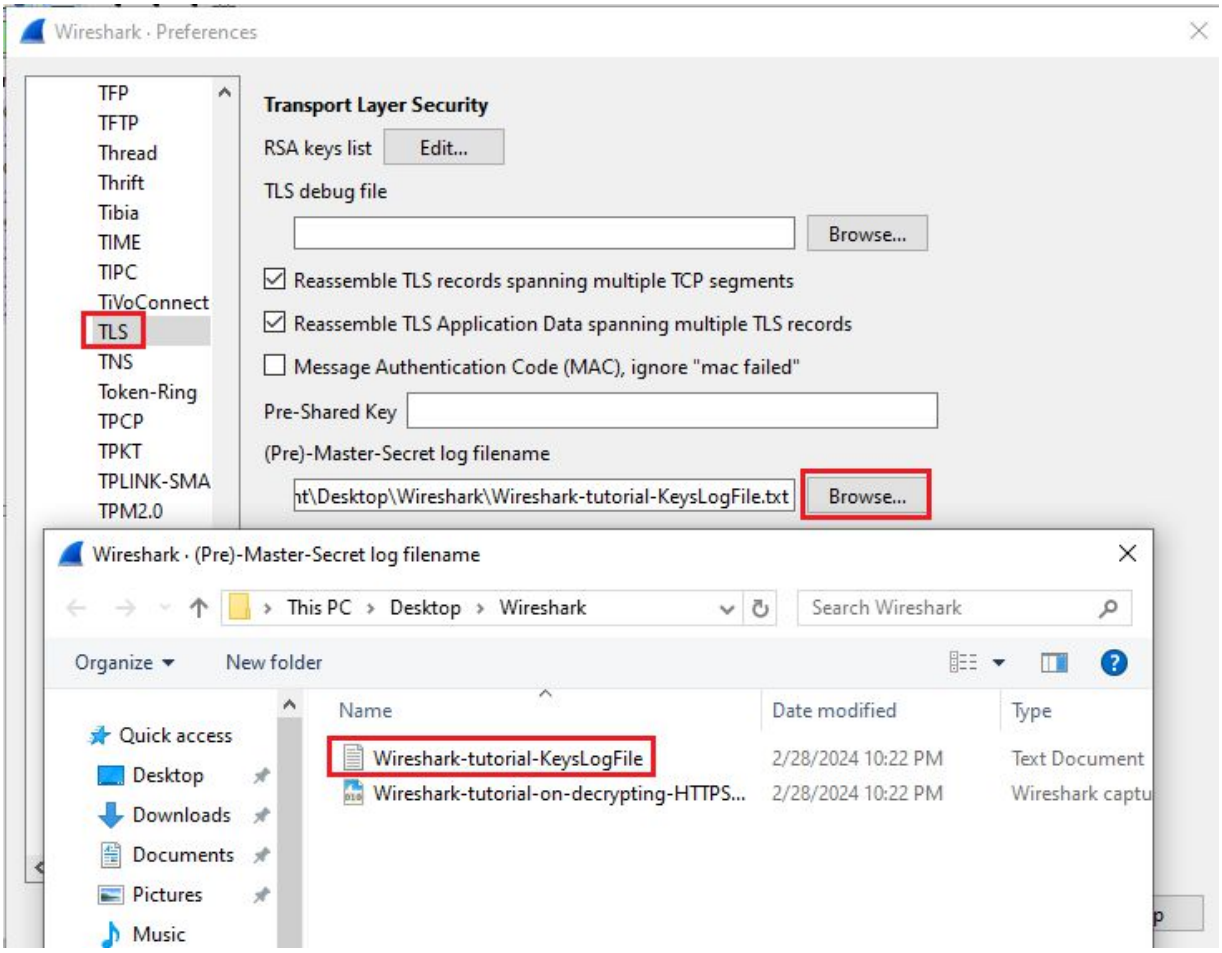
We import the decrypting key by clicking on Edit and scrolling to the bottom we click on Preferences.



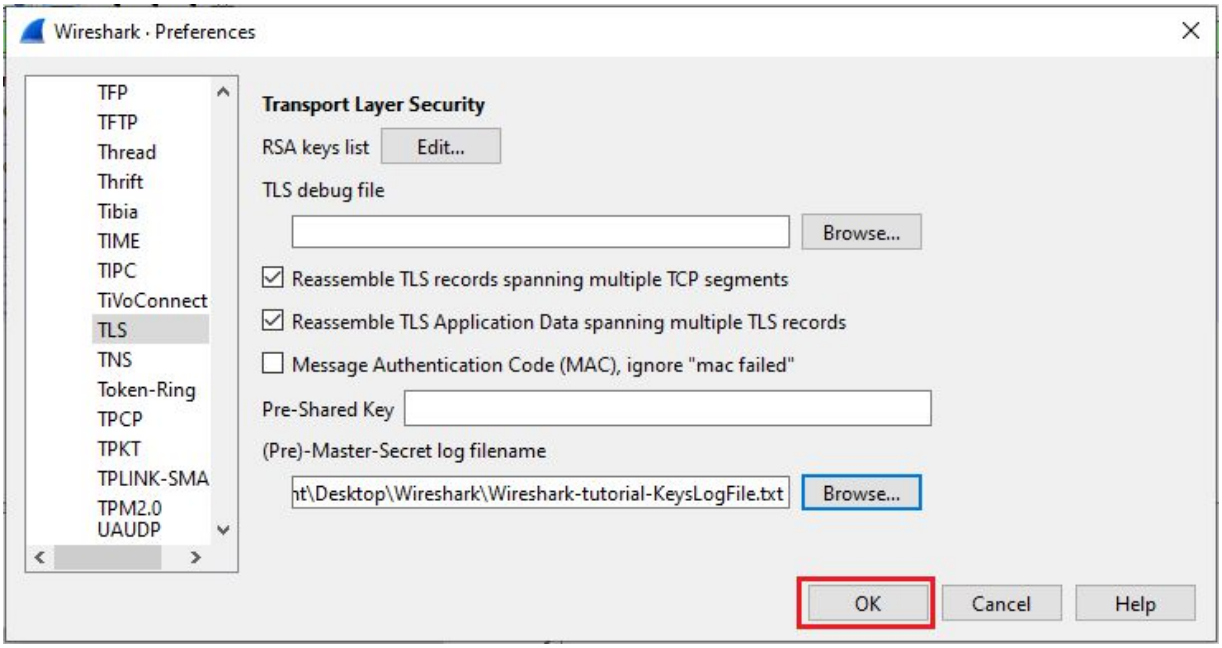
From the Preferences page, we click on Protocols. Scroll through the list until you come to TLS.



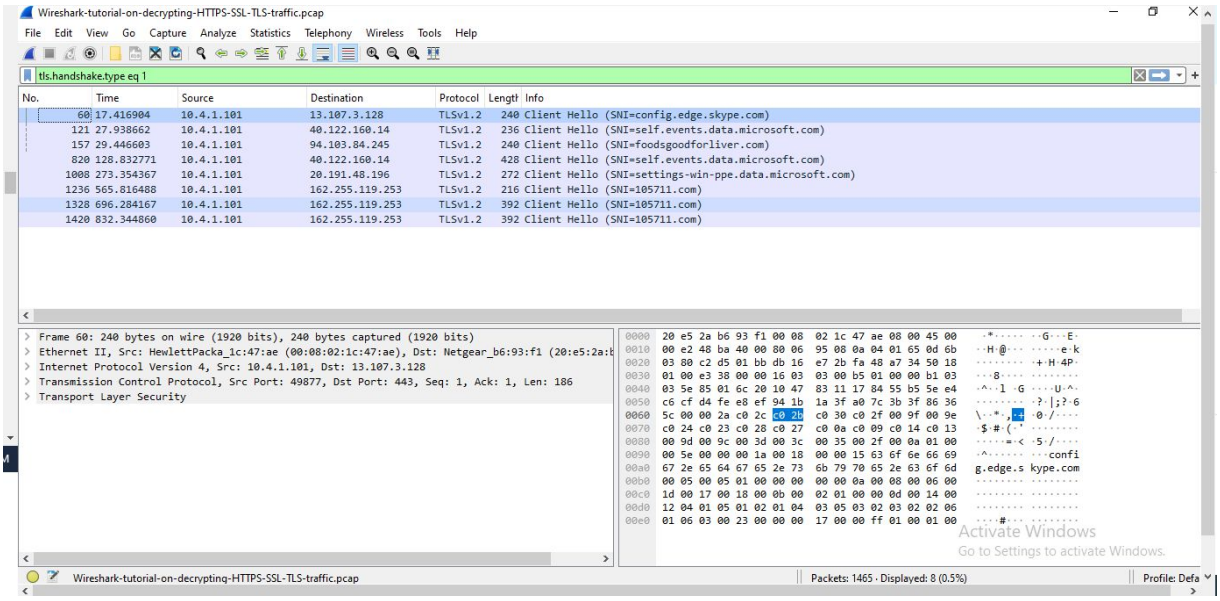
In the TLS properties window, click on the second browse button. Brown to the location oif your extracted archive and select the wireshark_tutorial_KeyLogFile.txt.


Wireshark · Preferences
TFP
TFTP
Thread
Thrift
Tibia
TIME
TIPC
TiVoConnect
TLS
TNS
Token-Ring
TPCP
TPKT
TPLINK-SMA
TPM2.0
Transport Layer Security
RSA keys list
Edit...
TLS debug file
Browse...
Reassemble TLS records spanning multiple TCP segments
Reassemble TLS Application Data spanning multiple TLS records
Message Authentication Code (MAC), ignore "mac failed"
Pre-Shared Key
(Pre)-Master-Secret log filename
ht\Desktop\Wireshark\Wireshark-tutorial-KeysLogFile.txt
Browse...
Wireshark · (Pre)-Master-Secret log filename
This PC › Desktop › Wireshark
Search Wireshark
Organize
New folder
Quick access
Desktop
Downloads
Documents
Pictures
Music
Name
Date modified
Type
Wireshark-tutorial-KeysLogFile
2/28/2024 10:22 PM
Text Document
Wireshark-tutorial-on-decrypting-HTTPS...
2/28/2024 10:22 PM
Wireshark captu

Click OK.


Wireshark · Preferences
TFP
TFTP
Thread
Thrift
Tibia
TIME
TIPC
TiVoConnect
TLS
TNS
Token-Ring
TPCP
TPKT
TPLINK-SMA
TPM2.0
UAUDP
Transport Layer Security
RSA keys list
Edit...
TLS debug file
Browse...
Reassemble TLS records spanning multiple TCP segments
Reassemble TLS Application Data spanning multiple TLS records
Message Authentication Code (MAC), ignore "mac failed"
Pre-Shared Key
(Pre)-Master-Secret log filename
ht\Desktop\Wireshark\Wireshark-tutorial-KeysLogFile.txt
Browse...
OK
Cancel
Help

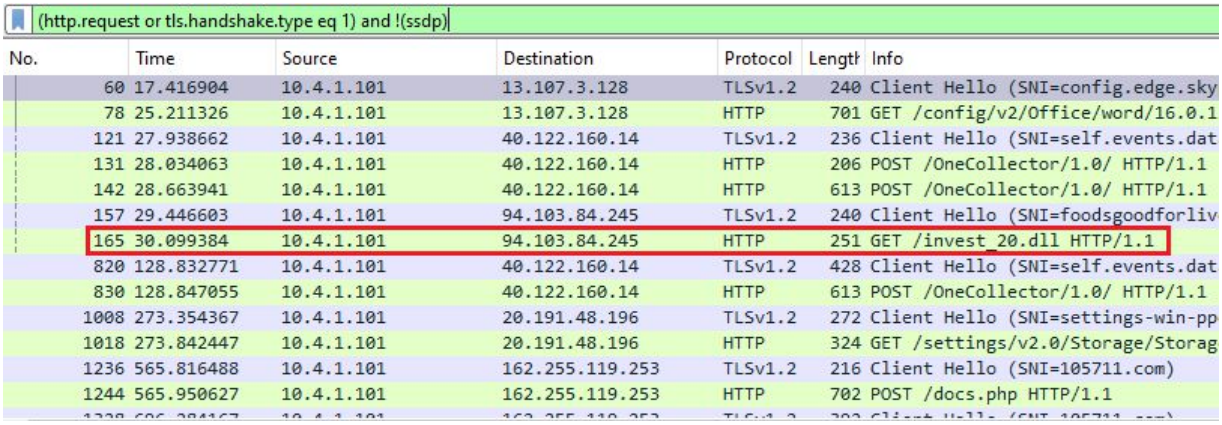
Once we click OK, the encrypted traffic is decrypted.



Analyzing the pcap file for malware.

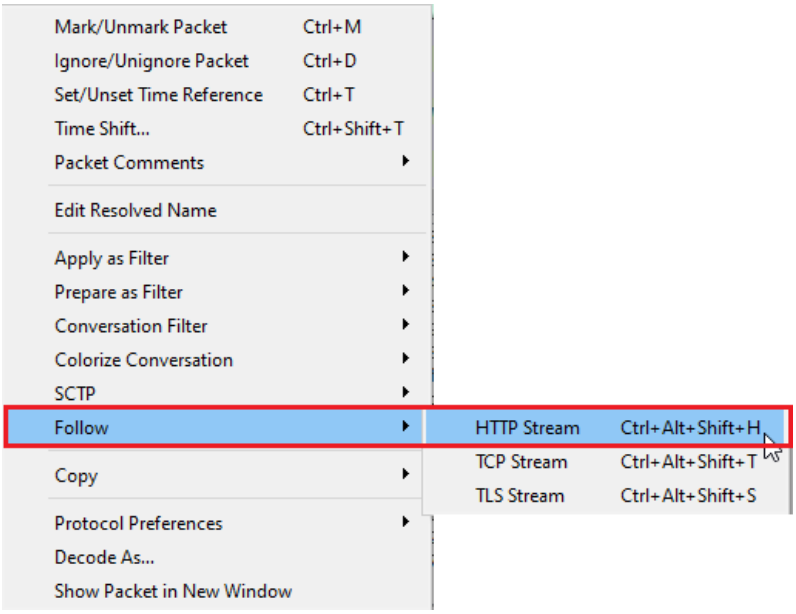
We next will want to filter for any HTTP TLS handshake traffic, excluding SSDP:

```
(http.request or tls.handshake.type eq 1) and !(ssdp)
```



On line 165, we see a GET request calling on a .dll file.

If we right-click on the GET packet, select and from the menu select Follow, and from the subsequent context menu select HTTP Stream, we can see that the .dll has already been downloaded.



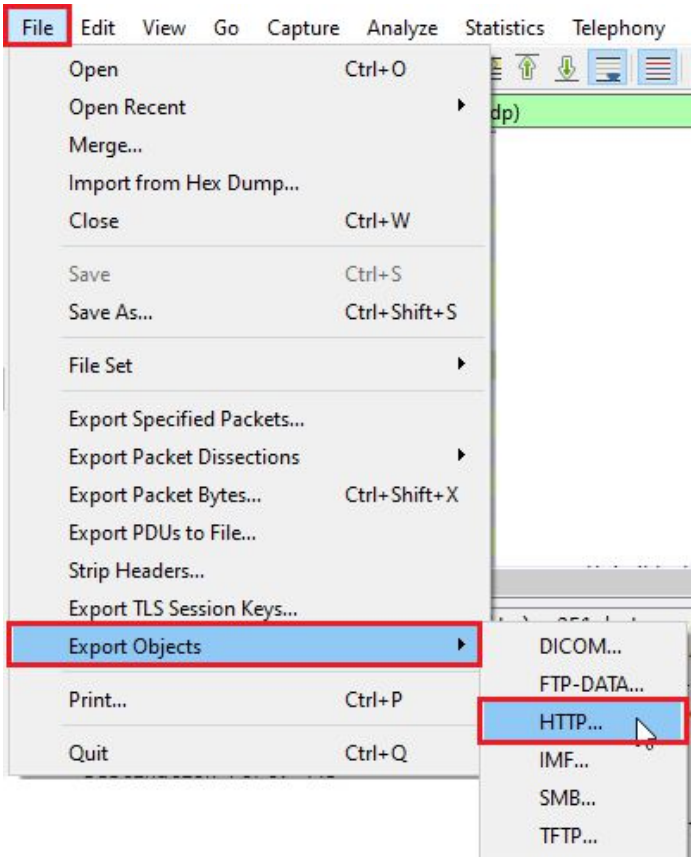
Everything below the line labeled, “This program cannot be run in DOS Mode,” is the .dll.

We can save the downloaded .dll and upload the saved file to a website like Virus Total to determine the malware type.

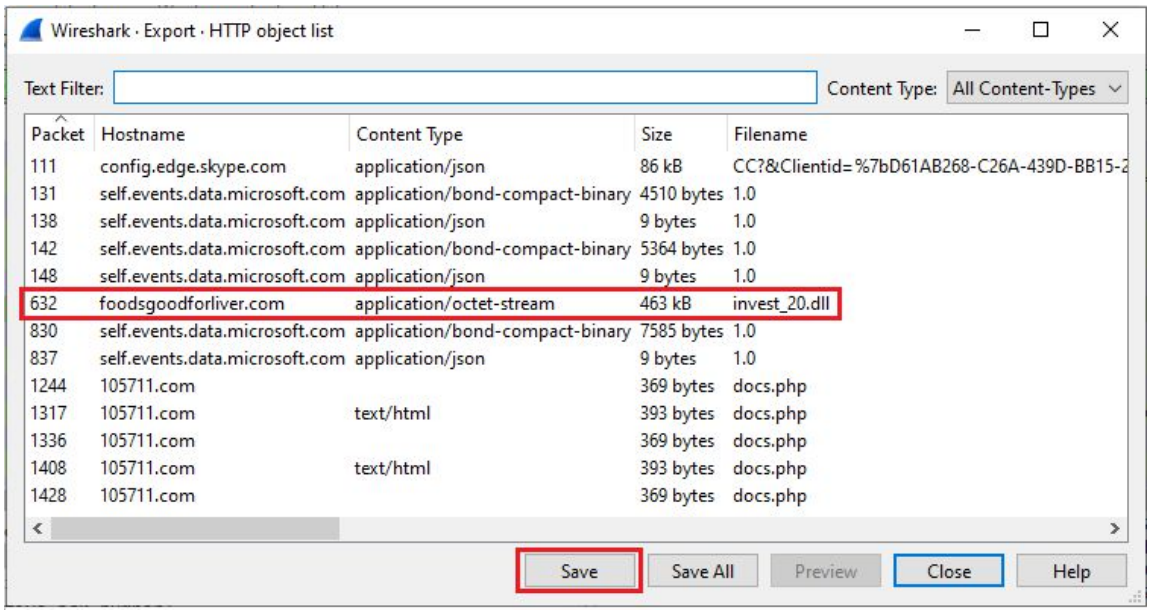
Download the DLL file.

Close out the HTTP Stream window. Back at the filtering windowpane, highlight line 165, showing the GET request for the .dll.

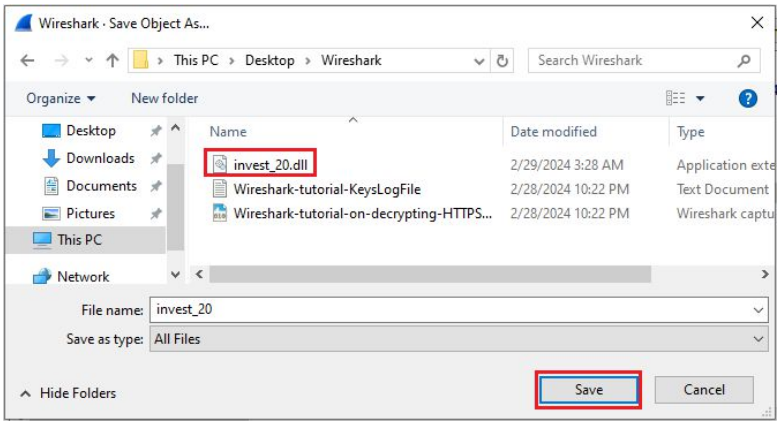
From the Wireshark taskbar, click File, Export Objects, and from the last content menu, select HTTP.



On the next screen, highlight the line containing the .dll you need to have analyzed and click save.

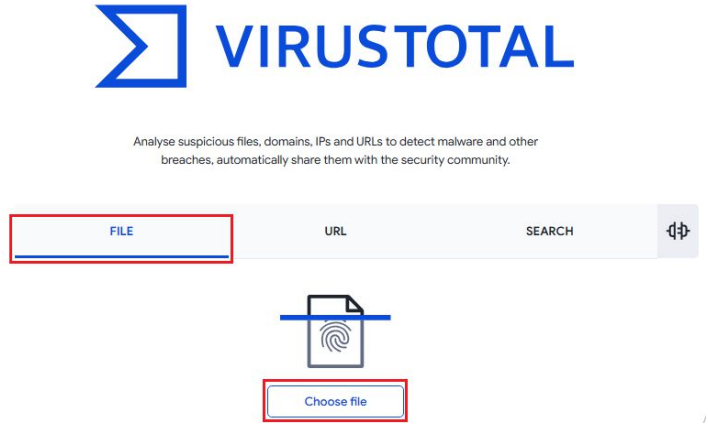


Save the file locally.

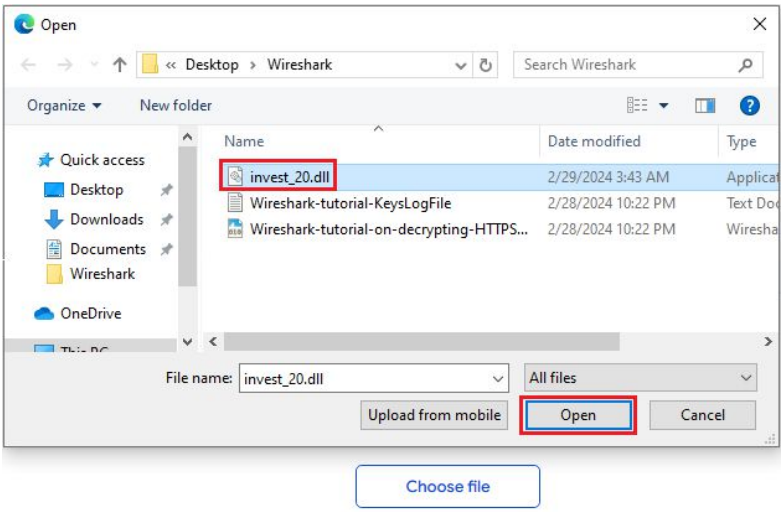


Open a browser, and in the address bar, go to virustotal.com.

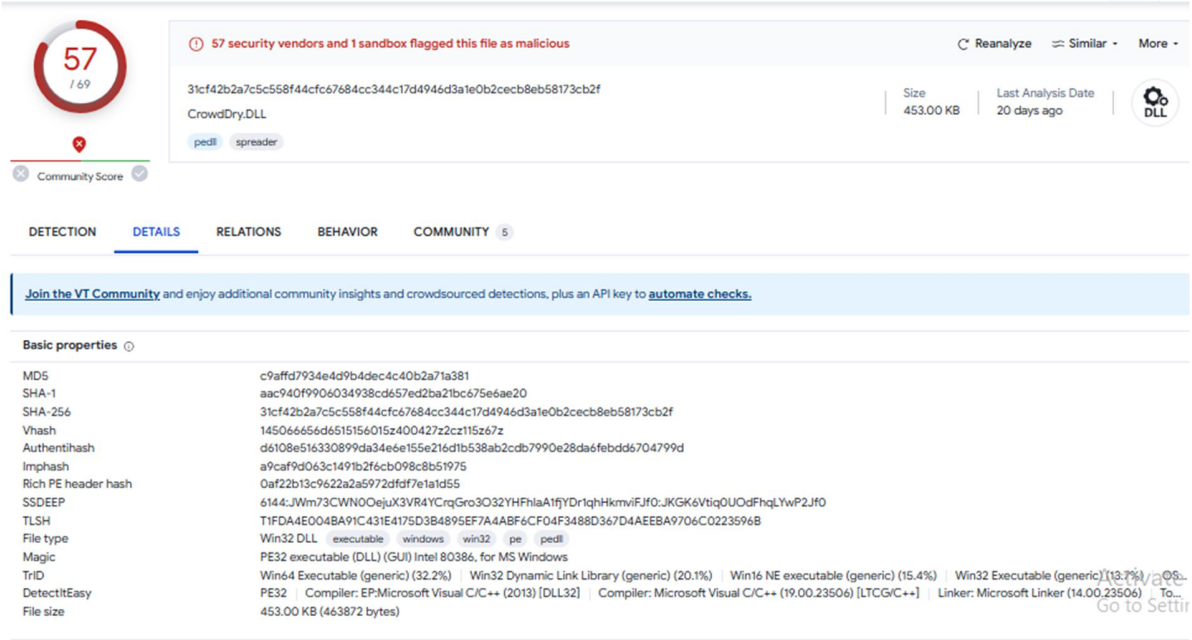
We are going to upload the saved file. Select the file option.



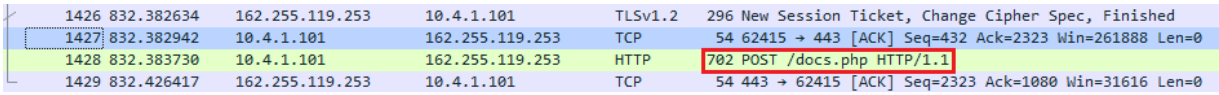
Next, click the option to choose the file.



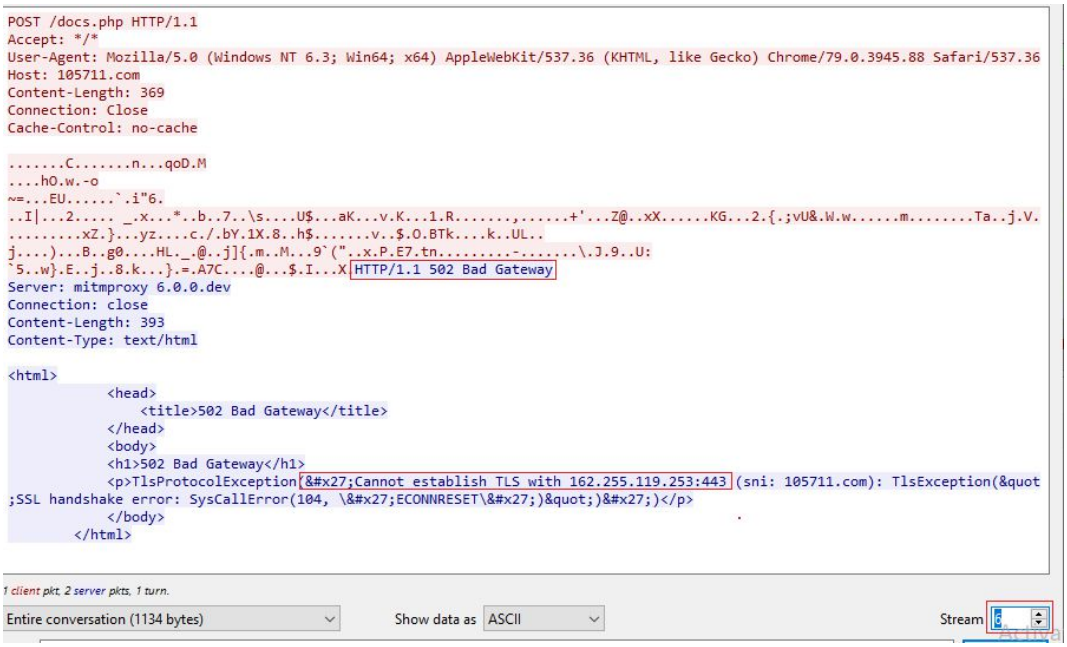
Virus Total gives us a very detailed breakdown of the file.

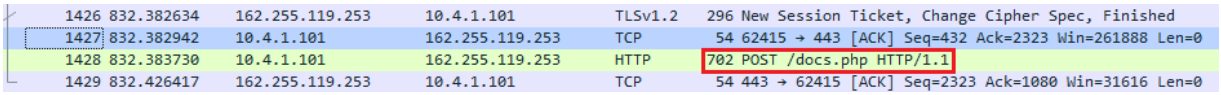


From our pcap results, we see a strange post request being initiated by a php file.

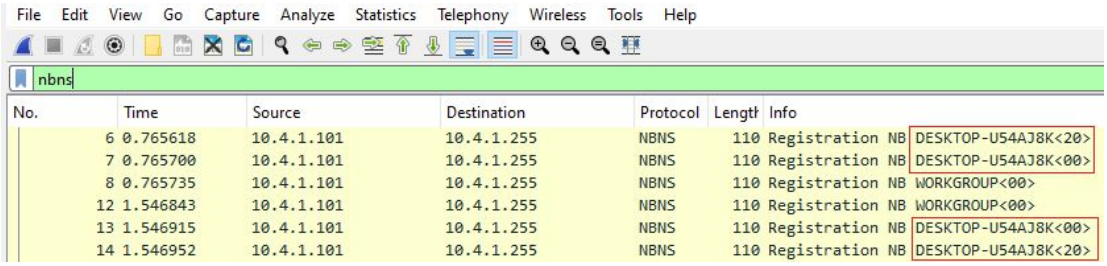


If we right-click on the packet, follow the TLS stream, and change the stream from 7 to 6, we can see that once the machine becomes infected, attempts to connect with a command-and-control server.





Lastly, we can identify the name of the machine by filtering for just the NetBIOS Name Service (NBNS).



Summary

In this forensic lab, we used Wireshark to scrutinize network traffic and uncover malicious activities. By leveraging Wireshark's capabilities, analysts were able to meticulously trace the malware's communication patterns, command and control infrastructure, and potential payloads. Through systematic examination of packet captures, this lab aimed to provide invaluable insights into detecting, mitigating, and preventing malware infections, thus fortifying network defenses against the threat of malware.